QP CODE: 24900105

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR

EXAMINATION NOVEMBER 2024
First Semester

Discipline Specific Core Course

ANSWER SCHEME

MG1DSCFSQ100 – INTRODUCTION TO FOOD SCIENCE

(2024 ADMISSION ONWARDS)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions.Each question carries 1 mark.

1.Recoganize one of the key benefits of refrigeration over earlier food preservation methods

Ansb)It slows down bacterial growth without significantly changing the taste or texture of food

2.Evaluate the main focus of Food Science

Ans c) Applying basic science and engineering to understand the physical, chemical, and biochemical nature of foods.

3.Select the correct type: A _____ typically sells goods in bulk to supermarkets or restaurants.

Ansc) Wholesaler

4.Identify the global food security index (GFSI)

Ans:a) Measures food security across countries.

5.Recognise the low risk food from given below

Ansb) Dry fruits

6.HACCP stands for

Ansd) Hazard analysis critical control point

7.Identify the main function of fiber in the diet

Ans:d) All of the above

8.Match the water activity value for the growth of yeast.

Ans: c) 0.8

9.**Name the elongated rod shaped bacteria**.

Ans:a) Bacilli

10. **Identify the natural food toxicant**

Ans : b) Phytotoxins

(10 × 1 = 10)

Part B

Short Essay Type Questions

Answer 4 questions. Each question carries 5 marks

11 Describe food science

Food science is a multidisciplinary field that focuses on solving real-world problems around the production of food. Modern consumers want safe, nutritious, and easily available food. Food scientists help farmers, grocers, and other businesses meet those demands.The study of food allows us to understand the plants and animals that we eat and how to improve current food systems to make them more efficient and sustainable.**(Definition 2 marks)**

The science of food involves studying:

Food composition

Preparation

Processing

Preservation

Effects on human health**(Description 3 marks)**

12 Define food safety and food hazards

Food Safety refers to handling, preparing and storing food in a way to best reduce the risk of individuals becoming sick from foodborne illnesses**.(1.5 mark)**

The principles of food safety aim to prevent food from becoming contaminated and causing food poisoning. It involves

- Properly cleaning and sanitising all surfaces, equipment and utensils
- Maintaining a high level of personal hygiene, especially hand-washing
- Storing, chilling and heating food correctly with regards to temperature, environment and equipment
- Implementing effective pest control
- Comprehending food allergies, food poisoning and food intolerance**(1mark**)

Food Hazard A food safety hazard is defined by FSIS HACCP as any biological, chemical, or physical property that may cause a food to be unsafe for human consumption**. (1.5 marks)**

.Five types of hazards that can affect how contaminants may potentially enter into the processed food: contact-surface hazards, overhead hazards, proximity hazards, potential transfer hazards and environmental hazards**.(1 mark)**

13 Analyse the effect of food safety controls in food industry

Food safety (or food hygiene) is used as a scientific method/discipline describing handling, preparation, and storage of food in ways that prevent foodborne illness. The occurrence of two or more cases of a similar illness resulting from the ingestion of a common food is known as a food-borne disease outbreak.

In considering industry-to-market practices, food safety considerations include the origins of food including the practices relating to food labeling, food hygiene, food additives and pesticide residues, as well as policies on biotechnology and food and guidelines for the management of governmental import and export inspection and certification systems for foods.

In considering market-to-consumer practices, the usual thought is that food ought to be safe in the market and the concern is safe delivery and preparation of the food for the consumer. Food safety, nutrition and food security are closely related. Unhealthy food creates a cycle of disease and malnutrition that affects infants and adults as well**. (5marks)**

14 As a food caterer point out the major things to keep in mind while cooking, handling and serving food products.

In the catering industry, ensuring food safety is paramount. Whether catering for a wedding, corporate event, or private party, maintaining high standards of food safety is crucial to the success of any catering service.

Maintaining a clean and sanitized environment is crucial in food preparation areas.

Catering services must follow strict sanitization protocols to prevent the spread of germs and bacteria. Surfaces, utensils, and equipment should be thoroughly cleaned and sanitized before and after each use. Frequent hand washing with soap and water is vital for all staff members, especially when handling food.

Additionally, gloves and hairnets are necessary to prevent contamination. Sanitization stations should be set up at various points in the food preparation area for easy access

Quality Ingredients (1 mark)

Using high-quality, fresh ingredients is essential for ensuring the food's safety and integrity. Catering services must source their ingredients from reputable suppliers and ensure they meet the highest standards of quality and freshness.

Proper storage and handling of ingredients are also critical in preventing contamination. Regular inspections of the ingredients and storage facilities must be carried out to guarantee adherence to safety regulations.

Temperature Control(1 mark)

Maintaining the correct food temperature is crucial in preventing the growth of harmful bacteria. Food not stored or cooked at the right temperature can quickly become a breeding ground for bacteria, leading to foodborne illnesses. Catering services must implement strict temperature control measures during food preparation and delivery. Using insulated containers and temperature-controlled equipment during transportation helps to ensure that the food remains at the correct temperature until it is served.

Contactless Delivery (1 mark)

In the current landscape, contactless delivery has become the norm. Catering services should offer contactless delivery options to minimize the risk of exposure to both clients and staff. This can include leaving the food at a designated location for pickup or using contactless delivery services. All delivery personnel should follow strict safety protocols, including wearing masks and gloves and maintaining social distancing guidelines.

Packaging and Sealing (1 mark)

Proper packaging and sealing of food are essential to prevent contamination during transportation. Catering services should use secure, sealed containers to package the food, minimizing the risk of exposure to external contaminants. Additionally, food labels should include important information such as the date and time of preparation, storage instructions, and allergen information, ensuring transparency and safety for all consumers.

Regular Testing and Monitoring (1 mark)

Routine testing and observation are crucial to guarantee that safety procedures work correctly and that any possible problems are quickly detected and resolved. Catering services should regularly inspect their facilities, equipment, and protocols to guarantee adherence to safety requirements. Additionally, food samples should be tested periodically to ensure they meet the required safety standards. Any deviations from the standard protocols should be addressed immediately to prevent any compromise in food safety

15 Discuss the socio cultural significance of food

Food has the remarkable ability to bring people together and create a sense of connection. It offers a unique opportunity to learn about different cultures and broaden our understanding of the world. By exploring diverse cuisines, we can gain insights into various societies' history, traditions, and social structures. For instance, analyzing the ways in which food is prepared and consumed can reveal significant information about gender roles and power dynamics in a given community

Since ancient times, food has been a cornerstone of cultures worldwide. It is essential for our survival, which is why preoccupation with food is a human universal that connects us all. Food practices are learned mainly through transmission from parents to children. The structure and timing of meals, where food can be consumed at table, with others, and how to eat eg, food manners, eating pleasure are learned from social groups eg, families, schools, and peers. The learning process is both explicit, such as verbal communication about what to eat and exposure to information and implicit, such as indaily routines structured for children8 and in modeling of food behaviour. Cuisine (food combinations, flavors, and seasonings) is a major influence in food acceptance and helps shape food preferences.Sharing food ensures the survival of the group both socially and materially. A companion is literally a person one eats bread with .On a day-to-day basis, food exchanges are crucial in maintaining good relations between individuals.

"Food … is not only a collection of products that can be used for statistical or nutritional studies. It is also a system of communication, a body of images, a protocol of usages, situations and behavior**(5 marks)**

16 Explain about the characteristics of successful entrepreneurs.

1. **Curiosity**
   Successful entrepreneurs have a distinct personality trait that sets them apart from other organizational leaders: a sense of curiosity. An entrepreneur's ability to remain curious allows them to continuously seek new opportunities. Rather than settling for what they think they know, entrepreneurs ask challenging questions and explore different avenues.**(0.5 marks**)
2. **Willingness to Experiment**
   Along with curiosity, entrepreneurs require an understanding of structured experimentation, such as design thinking. With each new opportunity, an entrepreneur must run tests to determine if it's worthwhile to pursue.**(0.5 marks)**
3. **Adaptability**
   Entrepreneurship is a process, and new challenges and opportunities present themselves at every turn. It's nearly impossible to be prepared for every scenario, but successful business leaders must be adaptable.**(0.5 marks)**
4. **Decisiveness**
   To be successful, an entrepreneur has to make difficult decisions and stand by them. As a leader, they're responsible for guiding the trajectory of their business, including every aspect from funding and strategy to resource allocation.**(0.5 marks)**
5. **Self-Awareness**
   A great entrepreneur is aware of their strengths and weaknesses. Rather than letting shortcomings hold them back, they build well-rounded teams that complement their abilities**. (0.5 marks**)
6. **Risk Tolerance**

Entrepreneurship is often associated with risk. While it's true that launching a venture requires an entrepreneur to take risks, they also need to take steps to minimize it.**(0.5 marks)**

7. **Comfort with Failure**
   In addition to risk-management and calculated decision-making, entrepreneurship requires a certain level of comfort with failure**. (0.5 marks)**
8. **Persistence**
   While many successful entrepreneurs are comfortable with the possibility of failing, it doesn't mean they give up easily. Rather, they see failure as an opportunity to learn and grow.**(0.5**marks)
9. **Innovative Thinking**
   While innovation in business can be defined as an idea that's both novel and useful, it doesn't always involve creating an entirely new product or service. Some of the most successful startups have taken existing products or services and drastically improved them to meet the changing needs of the market**. (0.5 marks)**
10. **Long-Term Focus**
    Most people associate entrepreneurship with starting a business. While the early stages of launching a venture, such as securing funding, are critical to its success, the process doesn't end once the business is operational**.**Entrepreneurship is a long-term endeavor, and entrepreneurs must focus on the process from beginning to end to ensure long-term success**.(0.5 marks)**

(4 × 5 = 20)

Part C

Essay Type Questions

Answer 2 questions. Each question carries 10 marks

17. Summarize quality control and quality assurance .

Quality Assurancerefers to a formalized system of review that takes place during a specific time period before submittal, designed to catch and correct any last-minute issues.

Quality Control , on the other hand, encompasses the entire life of the project and is an ongoing process, designed to address problems as they pop up.**(Definitions 4marks)**

Quality Assurance	Quality Control
Provides planned system of review and spot-check by senior staff	Provides complete 100% check of documents, plans, etc.
One-step review process before each submittal	Detailed process for each submittal with multiple steps
Occurs after QC is complete, but before submission	Occurs throughout the life of the project
QA review frequency is per the Quality Management Plan (QMP)	QC review occurs on all projects and submittals

Quality Assurance (QA) activities comprise a planned system of review procedures conducted by people not directly involved in the design compilation or development process.

Quality Control (QC) is a system of routine technical activities to measure and control the quality of the product as it is being developed. The QC process is designed to provide routine and consistent checks of the design and plan details, to identify errors and omissions which can then be addressed. Quality Control roles include:

Quality assurance (QA) is primarily concerned with preventing defects throughout the production process, emphasising proactive measures such as process improvement, training, documentation, and audits. On the other hand, quality control (QC) is focused on identifying and rectifying defects that may exist in the final product, emphasising inspections and testing to ensure product compliance.

,A QA manager's responsibilities may encompass a broader range of tasks, including hygiene, pest control, defect identification, regulatory compliance, and addressing customer complaints. Because quality assurance and quality control are two distinct but related functions, both are necessary in a food processing plant.

Difference between quality control and quality assurance?

Because quality assurance and quality control are two distinct but related functions, both are necessary in a food processing plant.

What does quality control do?

Quality control (QC) plays a vital role in upholding and enhancing product quality within a business. Its primary objective is to maintain consistency and improve product quality. QC accomplishes this by conducting rigorous testing and inspections on individual units to determine if they meet the specified criteria for the final product. In essence, quality control ensures that products adhere to the highest standards set by the company.

Role of quality assurance in the food industry

The importance of a good quality assurance system for food processors cannot be overstated. While we have outlined above what it aims to achieve and how it differs from quality control, there is still a broader role of quality assurance in the food industry.

Not only does the role of quality assurance in the food industry directly relate to food safety, which we can all agree is fundamental, it also directly contributes to business success and profitability.

The role of quality assurance in the food industry also covers compliance with legal, customer, internal/external standards, as well as continuous improvement of quality, record keeping, reporting, problem detection and solving, which ultimately ensures a safe, high quality product is delivered to the consumer.

The consistent production of high-quality products by food companies builds consumer trust, which builds brand loyalty, and this can help to keep you ahead of the competition.(**Differentiation 6 marks)**

18 How do food technologists contribute to improving food safety through process optimization and innovation.

Food technologists ensure the quality of food as it travels from farm to fork, applying food science to the selection, preservation, processing, packaging, distribution, and use of foods.Food safety is a paramount concern for individuals, communities, and nations around the world. The field of food science plays a crucial role in safeguarding our plates by employing scientific knowledge and principles to prevent food borne diseases .

Identifying and Analyzing Food Hazards

Food scientists meticulously identify and analyze potential hazards in the food supply chain. They study microbiology, chemistry, and toxicology to understand the nature and behavior of pathogens, contaminants, and other harmful substances that can compromise food safety. By employing advanced techniques and tools, such as DNA testing, mass spectrometry, and chromatography, they can detect and quantify these hazards accurately.

POINTS TO BE INCLUDED AND THEIR EXPANSION

- ✓ Implementing Good Manufacturing Practices (GMPs) and Standard Operating Procedures (SOPs.
- ✓ Developing and Implementing Food Safety Management Systems
- ✓ Evaluating Food Packaging Materials
- ✓ Monitoring and Controlling Foodborne Pathogens
- ✓ Assessing Allergenic and Intolerant Substances
- ✓ .Enhancing Food Safety Training and Education
- ✓ Conducting Risk Assessments and Regulatory Compliance
- ✓ Investigating Outbreaks and Tracing Contaminated Food **(10 marks)**

19 Describe the components of Food safety management system

A Food Safety Management System (FSMS) is a comprehensive framework designed to ensure food safety throughout the food supply chain. It consists of several key components that work together to identify, evaluate, and control food safety hazards. Here's an overview of its main components:

1. Prerequisite Programs (PRPs):

These are the basic conditions and activities necessary to maintain a hygienic environment and support food safety. Examples include:

Good Manufacturing Practices (GMPs): Guidelines for production, storage, and handling. Sanitation Standard Operating Procedures (SSOPs): Cleaning protocols to prevent contamination. Pest Control Programs Systems to prevent infestations.Personal Hygiene Policies: Rules for worker hygiene.

2. Hazard Analysis and Critical Control Points (HACCP):

- A systematic approach to identify, assess, and control food safety hazards. It involves:
- Conducting a hazard analysis.
- Determining critical control points (CCPs).
- Establishing critical limits for each CCP.
- Monitoring procedures.
- Corrective actions.
- Verification and validation.
- Record-keeping and documentation.

3. Management Commitment and Responsibility:

- Leadership must demonstrate commitment to food safety.
- Establish a food safety policy.
- Allocate resources for implementation and maintenance of the FSMS.

4. Risk Assessment and Hazard Identification:

- Identify potential biological, chemical, and physical hazards.
- Assess risks and implement controls to mitigate them.

5. Traceability System:

- A mechanism to track food products through all stages of production, processing, and distribution.
- Ensures quick identification and recall of contaminated products if necessary.

6. Documentation and Record-Keeping:

- Create and maintain documents such as policies, procedures, and records of food safety activities.
- Examples include logs for temperature monitoring, cleaning schedules, and supplier approvals.

7. Training and Competence:

- Regular training programs for employees on food safety standards, procedures, and practices.
- Ensure staff are competent to perform their roles effectively.

8. Internal and External Audits:

- Regular audits to verify that the FSMS is implemented and functioning as intended.
- Internal audits conducted by the organization; external audits by third-party certifiers or regulators.

9. Communication:

- Internal communication: Ensure all employees understand their roles in food safety.
- External communication: Engage with suppliers, customers, and regulatory authorities.

10. Corrective and Preventive Actions (CAPA)

- Identify and address non-conformities or deviations.
- Implement measures to prevent recurrence.

11. Continuous Improvement:

- Regular review of the FSMS to incorporate updates, address new hazards, and improve processes.

12. Emergency Preparedness and Response:

- Plans to address food safety emergencies, such as contamination incidents or recalls**.(10 marks)**

20 Discuss the functions of food

Ans: 1. PHYSIOLOGICAL FUNCTIONS OF FOOD

Energy Provision

Calories: Food provides energy required for daily activities, whether basic metabolic functions (like breathing and maintaining body temperature) or physical exertion.

Macronutrients: Carbohydrates, fats, and proteins serve as the body's primary energy sources, with carbohydrates being the most immediate and fats the most concentrated energy source**.**

2. Growth and Development

Proteins: Provide building blocks (amino acids) necessary for cell repair, tissue growth, and muscle development.

Micronutrients: Minerals like calcium and phosphorus are vital for bone formation, while vitamins like vitamin D support growth.

3. Repair and Maintenance

Food helps the body repair damaged tissues and maintain its functions.

Nutrients like proteins and certain vitamins (e.g., vitamin C for collagen production) contribute to wound healing and overall tissue repair.

4. Regulation of Body Processes

Hormone Production: Nutrients like iodine (for thyroid hormones) and fats (for steroid hormones) are critical for hormone synthesis.

Enzyme Function: Vitamins and minerals act as coenzymes and cofactors that help enzymes catalyze essential biochemical reactions.

5. Immune Support

Micronutrients: Vitamins (A, C, D, E) and minerals (zinc, selenium) play roles in strengthening the immune system.

Phytonutrients: Found in plant-based foods, these compounds help fight inflammation and oxidative stress.

6. Protection Against Disease

Antioxidants: Foods like fruits, vegetables, and whole grains contain antioxidants that protect cells from damage caused by free radicals, reducing the risk of chronic diseases like cancer and heart disease.

Fiber: Found in plant-based foods, fiber supports digestive health and reduces the risk of conditions like diabetes and colorectal cancer.

7. Psychological and Emotional Well-Being

Mood Regulation: Certain foods, such as those rich in omega-3 fatty acids or amino acids like tryptophan, are linked to mood enhancement and mental health.

Comfort and Social Connection: Eating is often tied to cultural practices, celebrations, and emotional comfort, making it a central aspect of social and emotional well-being.

8. Hydration

Many foods, especially fruits and vegetables, have high water content, contributing to the body's hydration needs.

9. Support for Specific Bodily Functions

Brain Function: Omega-3 fatty acids, glucose, and antioxidants support cognitive processes.

Muscle Function: Nutrients like potassium, magnesium, and calcium are essential for muscle contraction and relaxation.

Understanding the multifaceted roles of food highlights its importance not just for survival but for optimizing health, performance, and quality of life**.**

SOCIAL FUNCTIONS OF FOOD

Food plays a central role in social life and serves a variety of social functions that go beyond its basic purpose of nourishment.

1.Expression of Culture and Identity Immunity Building and Social Bonding

2. Hospitality and Generosity

3. Facilitating Communication

4. Social Norms and Inclusion

5. Emotional Comfort and Memory Sharing

PSYCHOLOGICAL FUNCTIONS OF FOOD

The psychological functions of food are tied to its impact on mental and emotional well-being. These functions include:

Comfort and Stress Relief: Food provides emotional comfort, often serving as a coping mechanism during stress, sadness, or anxiety (e.g., "comfort food").

Social Connection: Sharing meals fosters social bonding, cultural traditions, and a sense of belonging, reinforcing psychological well-being.

Mood Regulation: Nutrients like carbohydrates, proteins, and certain vitamins influence brain chemicals like serotonin and dopamine, affecting mood and emotional states.

Identity and Self-Expression: Food choices can reflect personal or cultural identity, enabling individuals to express values, preferences, and heritage.

Reward and Motivation: Eating pleasurable foods activates the brain's reward system, reinforcing behaviors and creating feelings of satisfaction.

Memory and Nostalgia: Foods can evoke memories and emotions associated with past experiences, offering psychological comfort and connection to one's history.

Mindful Eating and Relaxation: The act of eating mindfully can promote relaxation, reduce stress, and enhance the connection between body and mind**.(!0 marks)**

(2 × 10 = 20)